



Pokhara University
Everest Engineering College
Final Internal Assessment
Spring- 2025

Level: Bachelor

Program: BE CMP(6th Semester)

Faculty: Science & Technology

Section:

Subject: Computer Network

F.M. 100

P.M. 45

Time: 3hrs

Attempt all the questions.

- 1 a) Explain how client server architecture is more secure than peer to peer networks? 7
- b) Discuss the Seven Layers of OSI protocol stack. Also compare TCP/IP and OSI layers. 8
- 2 a) Discuss how packet switching is different from circuit switching 7
- b) Differentiate between error detection and error correction. Describe how Hamming code works for data of 7 bits. Show that it can detect single bit error using an example. 8
- 3 a) Explain IPV4 header format. Differentiate between IPv4 and IPV6 7
- b) The APNIC Pool for Pokhara University (103.16.32.0/22) is to be divided into network of 7 different schools. Among seven schools three school need to be subdivided into two different departments. Provide a complete IP address plan which include network address, broadcast address, usable IP pool, subnet masks and wildcard masks 8
- 4 a) Why congestion occurs in the network? Explain the type of closed loop congestion. 8
- b) Define DHCP. Explain the iterative and recursive DNS query for name resolution. 7

- 5 a) What is Adaptive and Non Adaptive Routing? State and explain Link state Routing with example. 7
- b) Explain the working Principle of RSA algorithm. If $N=119$, public key $E=5$, and private key $D=77$, then demonstrate how to end the character plain text $PT=6$ using RSA 8
- 6 a) Discuss VPN and its types. 7
- b) What are the different guided media explain in brief 8
- 7 Write short notes on: (Any two) $2 \times 5 = 10$
- a) Proxy Server
 - b) Frame Relay
 - c) SMTP

*****Best Wishes*****